

Meta-Data and RGB Images: DS9 Part II

You should work together with a partner on all in class activities and have discussions, but each person should work on their own computer, and complete their own worksheet.

*Specific items requiring a response are in **bold red font**. I suggest making your own document where you record those answers, which you will upload when completed.*

Credit: This activity is adapted from “A Beginner’s Guide to Working with Astronomy Data” (BG), S3.4-3.5

Learning Objectives: Be able to inspect and extract meta data information (fits header).
Combine multiple wavelength images to make a false color RGB image.

We will pick up where we left off at the end of last class. Astronomical images can only be interpreted correctly if you know the conditions under which they were produced. FITS files include this meta-information (what the image is, its coordinates, its exposure time, etc.) in a dedicated section called the header.

A. Inspecting Data Meta-Information

1. **Load the data:** Open DS9 and load the WFPC2 Eagle Nebula data file we worked with last class.

2. **Read observation information:** Inspect the FITS header of the Eagle Nebula image by going to **F i l e** in the main menu, and choosing **H e a d e r**. Once you do this, you are given two choices; the second one is the file name with [SCI] appended. Start by looking at the first header. You don’t need to understand this in detail (Neither I or the BG author do!), but some of the items are self-explanatory. What does this header tell you about the following (you will need to scroll down):

- **Date the image was taken:**
- **Name of the telescope:**
- **Instrument taking the data:**
- **Position of the Sun at the moment of observation (RA_SUN, DEC_SUN):**

- **Relative angle to the position of the moon (MOONANGLE):**
- **Observation start time and end time (EXPSTART , EXPEND):**
- **What grant supported this observation (HISTORY):**

3. **Reading data information:** Close this header and open the second one, which has the file name with [SCI] appended. This is more image specific. What does this header tell you about the following:

- **The number of axes in the data (NAXIS):**
- **The width and height of the image in pixels (NAXIS1 , NAXIS2):**
- **The units of the pixel values (BUNIT):**
- **The degrees that correspond to the width and height of a single pixel in degrees (CD1_1 , CD2_2):**

Note: The CRVAL1 , CRVAL2 , CRPIX1 , CRPIX2 and the CD1_1 , CD1_2 , CD2_1 and CD2_2 contain the information that allows you to calculate, for each pixel, the values of the two equatorial coordinates: right ascension and declination. The CD values correspond to a matrix mapping the two coordinate systems to each other. Often, the axes of the WCS and the image coordinate systems are parallel to each other.

You can check this in the overview window in the upper right corner, second from the right, where the North and East directions are shown as N and E, and the image coordinate directions as X and Y — common convention is to create images where North is up and East is left, as it would be if you were facing North and looking at the night sky. If your image is not like that, you can click Frame - Match Frames - WCS.

In that case, CD1_2 and CD2_1 are zero, and for the usual square pixels, CD1_1 and CD2_2 are equal (up to a sign) and denote the image scale. CD1_1 has an extra minus sign because the X direction and the East direction are anti-parallel in this image.

B. Estimating Physical Size

4. **Calculate the image size:** Use the the information to measure the image size.

- **How many arc seconds does the width of the pixel correspond to? Recall: 1 degree = 60 minutes; 1 minute = 60 arc seconds.**
- **What is the width and height of our image in degrees?**
- **If the distance to the Eagle Nebula is 1.74 ± 0.13 kpc, what is the true physical size of the image? Show your work!**

Finished already? Download and start Homework #2